

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

GET 206 — Past Questions & Materials
by Renance

Course: GET 206
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

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Paper 1 of 3 — GET 206 - 2024/2025 - Second Semester - BASIC THERMODYNAMICS

Level 200 • Second Semester • PastQuestion

COLLEGE OF ENGINEERING AND

: DEPARTMENT OF MECHANICAL EN $\text{\AA}$

SECOND SEMESTER EXAMINATION 202

COURSE CODE AND TITLE: GET 206 - BASIC THERMODY!

) TIME ALLOTTED: TWO (2) HOURS

go INSTRUCTION: READ THE QUESTIONS CAREFULLY BEFORE ANSWERING.

(THE INPUT VALVE, "x" IS THE MIDDLE PART O R

QUESTION 1

(a) A young researcher claims to have invented a machine with an efficiency of 100 %. Using your k
NS First Law of Thermodynamics, briefly refute or support this claim. :

ra x (b) Consider a water-cooled condenser in a large refrigerator system in which R-134a is the
refrigerant fl

& The refrigerant caters the condenser at 1.0 MPa and 60 °C at the rate of 0.2 kg/s and exits as a
liquid at

29.05 MPa at 35 € Cooling water enters the condenser at 10 °C and exits at 20 °C. Determine the
rate at

Kod

grt y> which cooling water flows through the condenser. (Solve to three decimal places), Take:
R-134a inlet

« enthalpy = {x/30} kJ/kg: R-134a liquid outlet enthalpy = {x/45}; Water inlet enthalpy = {x/2B6}
kJ/kg:

yes.

water outlet enthalpy = {x/150}

» (c) Students of Chemical Engineering had designed an insulated chamber for Thermofluids
studies, Calculate

aon: enthalpy of the output stream, if the insulated chamber has two inputs streams, h_i and i_p and
one output

stream, m . The first input stream has a mags of 8kg/s and a specific enthalpy of 10kJ/kg; while the :
2 second input stream has a mass of 18 kg/s with enthalpy of 12.1kd/kg. an

ot QUESTION 2 yw fun

A (a) (i) State the Second Law of Thermodynamics and its consequences

(b) The Minister of Petroleum Resources recently approved FUPRE's request for supply of crude oil
to run its

4 NY modular refinery developed by the University's Lecturers, A crude oil depot and pump were
designed and

a constructed io temporarily store the crude and pump same to the refinery, respectively, The
pressure

us os, differential was estimated to be 1 bay, while the specific volume was calculated to be 0,001 m³/kg.

Calculate the pump work

(c) (i) Two heat engines are connected in series, heat engine 1 is the heat source for heat engine 2. If the F

efficiency of the first heat engine is 30% and the second heat engine is 35%, calculate the overall efficiency

for an irreversible heat engine? (ii) if the work done by the first engine is 68.5, and the heat

put into the second heat engine: Take the overall efficiency to be 47 %

QUESTIONS

(a) A small throttling valve was designed by the Centre for Offshore and Maritime Studies, FUPRE. In this

small throttling valve, an ideal gas was undergoing throttling. Assuming negligible changes in potential and

kinetic energies (i) write an expression relating changes in specific enthalpy, h , AT and u (ii) write an

expression relating initial pressure, P_0 , initial specific volume, v_0 and final pressure, P_1 , final specific volume, v_1

(b) To study a novel corrosion case brought to The Centre for Corrosion Studies FUPRE, by TotalEnergies, a

non-insulated tank having initial mass of not equal to zero ($m \neq 0$) was designed. A perfect gas was used to

charge the control volume, that is the non-insulated tank. The mass at state point 1, m_1 , temperature, T_1 are

0.2 kg and 25°C , respectively; while the mass at state point 2, m_2 , volume of the perfect gas, V_2 ,

pressure of the perfect gas, P_2 are 0.1 kg, 0.5 m³ and 10 Pa respectively. Heat added to the control volume,

is 8 J and the ratio of specific heats, $\gamma = 1.4$ (Solve to three decimal places).

QUESTION 3) Consider a system having work of 20 kJ and heat of 25 kJ interactions at its boundary,

calculate the resultant change in energy of your system? At $Q = 25$ kJ

QUESTION 4

(a) Consider Lecture Theatre, LT1 as a control volume filled with perfect gas undergoing a polytropic process,

Draw a schematic diagram of the system clearly labelling the following: surroundings, system boundaries,

energy interactions and mass interaction.

(b) Consider a control volume having fluid subjected to internally reversible steady flow. In this flow, air of

mass 5 kg is flowing at a velocity of 10 m/s and at a height of 1.5 m from the reference point. Calculate

changes in kinetic energy and potential energy. 250 UB St

O19 (c) A 5.0 litre tank contains 2.0 moles of a real gas at a temperature of 300 K. Using the vanDer waals

equation, calculate the pressure of the gas if $a = 0.364 \text{ Jm}^3/\text{mol}^2$ and $b = 4.27 \times 10^{-5} \text{ m}^3/\text{mol}$

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Paper 2 of 3 — GET 206 Past Questions - Fundamentals of Thermodynamics

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PENANGIUM RESOURCES, EFFURUN

DEPARTMENT OF (AIGIIMINGICAL ENGINEERING

First Semester Examination @0D6/2017 Academic Session

COURSE CODE AND TITLE: MBE 216 - BASIC-THERMODYNAMICS

DATE: APRIL 1ST, 2017_ TIME ALLOWED: 2 HOURS

INSTRUCTIONS: ANSWER #INY FOUR (4) QUESTIONS.

QUESTION 1

(a) Mention any four (4) terms/statements essential for inclusion in a complete definition of Thermodynamics. [2 Marks] ‘

(b) Highlight one major difference between Classicz! Thermodynamics and Statistical Thermodynamics.

[2 Marks]

(c) Is it true the assertion: “all closed systems are adicbatic”? Justify your answer. [2 Marks]

(d) How many thermodynamic states would be contained in a thermodynamic cycle with five (5) distinct

processes? Lllustrate the cycle on a property (P-V) diagram, labeling clearly the states and processes.

[4 Marks]

(e) Could the mixture of liquid air and gaseous air be regarded as pure substance? Justify your answer.

{2 Marks]

(f) A temperature scale of certain thermometer is given by the relation $t = x \ln p + y$, where x and y are constants

and p is the thermometric property of the fluid in the thermometer. If at the ice point and steam point the

thermometric properties are found to be 2.5 and. 8.0 respectively, what will be the temperature corresponding to the thermometric property of 4.0 on Celsius scale? [5.5 Marks]

QUESTION 2

(a) Mention the basic forms in which energy moves through the boundaries of thermodynamic systems. [2

Marks]

(b) If a Linear Fresnel solar collector system releases 250 kW of heat, out of the total 280 kW it absorbs from

tice Sun, as part of the processes required to produce 40 kW of eleciricity. What are the values of thermal

ee ut and output from the solar collector system? [2 Marks]

(c) State the mathematical expression of the First law of Thermodynamics, as it applies to ion-flow processes.

[1 Mark]

(d) Consider an ideal thermal power plant, based on Organic Rankine Cycle (ORC), which operates with silicon oil,

If 200 kW of heat is used to evaporate the silicon oil inside the ORC, and the turbine component of the ORC

produces 160 kW of mechanical work; determine the value of internal energy capacity of the ORC system,

assuming it is closed. Identify the working substance of the system. [5 Marks]

(e) A diesel engine piston which has an area of $45 \times 10^{-7} \text{ m}^2$ moves 5 cm during part of suction stroke, 300×10^{-6}

m^3 of fresh air is drawn in from the atmosphere. The pressure in the cylinder during suction stroke is 0.9×10^5

Pa and the atmospheric pressure is $1.013 \times 10^5 \text{ N/m}^2$. The difference between the suction and atmospheric pressure is accounted for flow resistance in the suction pipe and inlet valve. Find the net work

done during the process, [7.5 Marks]

QUESTION 3

(a) A researcher asserts that he has developed a thermal engine with an efficiency of 100%. With your

knowledge of the laws of thermodynamics, give your criticisms of this engine. [4 marks]

(b) State the difference between an isolated process and an adiabatic process [2.5 marks]

(c) State the cold air standard assumptions. [2.5 marks]

(d) A group of FUPRE research students carried out an analysis of the Carnot cycle. In this analysis, these young and

~Giifany minds gave the following information: that the Carnot cycle operates between 600°C and 50°C , and it

produces 130 kJ of work. Another group of students from Uniben determined, using the information provided by

FUPRE research students, the following: that (i) the thermal efficiency of the Carnot engine analyzed by FUPRE

students is 53.5% (ii) the heat added is 243 kJ and (iii) entropy change during heat rejection process is

0.361 kJ/K . Show by detailed calculations whether Uniben students are right or wrong? [8.5 marks]

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QUESTION 4

(a) What is (or are) the limitations and criticisms of the first law of thermodynamics? [2 marks]

(b) Derive an expression for the thermal efficiency of an Otto cycle [4 marks]

(c) Why is the actual efficiency of a thermodynamic cycle always less than the ideal efficiency? [2 marks]

(d) 0.5kg of air (ideal gas) executes a Carnot power cycle having a thermal efficiency of 50%. The heat transfer to

the air during the isothermal expansion is 40kJ. At the beginning of the isothermal expansion the pressure is

7bar and the volume is 0.12m³. Determine: (i) the maximum and minimum temperatures for the cycle in Kelvin. (ii)

the volume at the end of the isothermal expansion in m³. (iii) the heat transfer for each of the four processes in

kJ. For air, take $C_v = 0.171$ kJ/kgK & $C_p = 1.008$ kJ/kgK [9.5 marks]

QUESTION 5

(a) With the aid of well labeled P-v and T-s diagram, describe how pressure, volume and temperature changes within

the Dual cycle. [4.5 marks]

(b) A diesel cycle has a compression ratio of 15 and heat addition at constant pressure takes place at 6% stroke.

Find the air standard efficiency of the engine, Take ratio of specific heats to be 1.4. [5.5 marks]

(c) List and explain the processes in a Carnot cycle. [4.5 marks]

(d) Derive an expression for the efficiency of a Carnot cycle. [4 marks]

QUESTION 6

(a) Very seasoned Engineers had strongly criticized Otto cycle and Diesel cycle, arguing that they are oversimplification of an attempt at solving complex problems. Based on this criticism, the Dual cycle was

developed. Clearly, but briefly, explain these limitations that led to the development of the Dual cycle. [5 marks]

(b) Show that the equation, $B = \frac{P}{\rho} + \frac{1}{2}u^2 + \frac{1}{\rho} \frac{dP}{dx}$

force; P is pressure, ρ is density, u is velocity, μ is viscosity and x is distance. [2 marks]

(c) Compare and contrast Work and Heat? [2 marks]

(d) From the figure below, where $V_e = 0.0048$ m³, $V_s = 0.0195$ m³, $\gamma = 1.4$ (i) what is the name of the cycle whose P-v

and T-s diagram are shown in the figure below. (ii) estimate the air standard efficiency (iii) mean effective

pressure is either dimensionally homogeneous or not. Where the unit of B, is

pressure γ work done $\frac{1}{\gamma} \frac{P}{\rho}$

; $v = \frac{1}{\rho}$: m³/kg

85bar - = 3, $V_e = 4$ ie as

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VeVs Wea" ¥ key: $r=(v_i/v_2) = (V_{st}V_e)/V_e$ ‘

— [8.5 marks]

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination 2017/2018 Session

COURSE CODE AND TITLE: MEE 216 - BASIC THERMODYNAMICS

INSTRUCTION: Answer question one (1) is compulsory and any other three questions. TIME

ALLOTTED: 2.5Hrs

Question 1 (25 marks)

(a) Define any five of following terms: (i) path function, (ii) superheated. vapour, (iii) critical point, (iv)

thermodynamic equilibrium, (v) perpetual motion machine of the first kind (PPM1), (vi) weak mixture, (vi)

refrigeration capacity, (vii) throttling process and (viii) coefficient of performance

(b) Highlight reason why the Carnot cycle is considered impracticable in real life.

(c) Itemize the basic laws upon which the study of the thermodynamics is based.

(d) A closed system changes from state 1 to state 2 while 40 kJ of heat is added and 60 kJ of work is done

by the system. As the system is returned to state 1, 35 kJ of work is done on it. What is the heat transfer during process 2-1?

(e) (i) Compare and contrast the difference between Carnot, Otto, diesel and actual cycles or (ii) distinguish

between intensive and extensive properties.

Question 2 (15 marks)

(a) Compare and contrast the difference between isothermal process and adiabatic process.

(b) With the aid of a labeled diagram, explain Carnot cycle.

(c) A gas expands reversibly according to the relation $PV^n = C$. The initial volume and pressure of the gas are

0.0283 m³ and 13.8 bar respectively. The final pressure and volume are 1.035 bar and 0.2073 m³. Calculate the polytropic index of expansion and determine the work done during the process.

~ Question 3 “ (15 marks)

(a) Distinguish between a compressor and a pump or a nozzle and a diffuser.

(b) Distinguish between a homogeneous system and a heterogeneous system.

(c) A heat engine rejects 600 kJ of heat energy as it does 400 kJ of work. What is the thermal efficiency of the heat engine.

“Question 4 (15 marks)

(a) Differentiate between closed system, open system and an isolated system.

(b) State three similarities and differences between work and heat.

(c) 0.92 kg of a certain perfect gas at a temperature of 150 °C and pressure of 6.25 bar is constrained in

a piston-cylinder assembly and occupies a volume of 0.185 m³. The gas is allowed to expand at constant pressure until the volume is 0.85 m³. Calculate (i) the molecular mass, (ii) the final temperature, and (iii) the work done by the gas. (Take $R_u = 8.314 \text{ kJ/Kmol/K}$)

Question 5 (15 marks)

(a) What is the usefulness of the ideal Rankine cycle or why is it introduced?

(b) Define the following; (i) pure substance, (ii) triple point, (iii) sub-cooled liquid and quality liquid

(c) (1) Draw the following process on the PV-plane and derive an expression for the work done on (i) polytrophic process, (ii) isobaric process and (iii) isothermal process OR (2) Distinguish between macroscopic and microscopic approaches. . cz

-Question 6 (15 marks) Wwe W: Fe preleg-con~

(a) How is evaporation different from boiling 7

(b) Comment on the statement “Energy is always conserved, but its quality is always neglected.

(c) A fluid undergoes a reversible adiabatic compression from 4 bar, 0.3 m³ to 0.08 m³, according to the law,

$p v^{1.25} = \text{constant}$. Determine (i) change in enthalpy; (ii) change in internal energy; (iii) heat transfer, and

(iv) work transfer -

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

First Semester Examination - 2025/2016 Academic Session

COURSE CODE AND TITLE: MEE 26 - BASIC THERMODYNAMICS

DATE: APRIL 19, 2026. TIME ALLOWED: 2 Hrs

“INSTRUCTIONS: -ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS.

- STUDENTS SHOULD BE ALLOWED TO USE THEIR THERMODYNAMIC PROPERTIES TABLES.

QUESTION 1 (COMPULSORY) — thermodynamics < Wee science “therm cart.

(i) Define Thermodynamics, “* e895 emd, m= esel Sycleme: ersten [2 Marks]

(ii) Summarize the basic laws upon which the study of Thermodynamics is based. Eh ee, [2 Marks]

(iii) - Mention the essential components required for complete definition of a thermodynamic system

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[3 Marks]

(iv) State one (1) each of the basic similarities and differences between Steady flow and Non-flow systems. . J [2 Marks]

(v) Mention the forms in which energy exists. List transition, clear mmd work - [2 Marks]

(vii) How many processes would be contained 'in a thermodynamic cycle with three @) die inct states:

Illustrate the cycle on a pressure. (P-V) diagram, indicate clearly the states and processes,

[4 Marks]

(viii) State the Joule's law of internal energy. (number of) of Ported gas 6m function [1 Mark]

(ix) List the four (4) types of equilibrium states that a thermodynamic system must achieve for it to attain a state of thermodynamic equilibrium. — "> essay cess cel rare Marks |

(x) - A closed thermodynamic system: contains air at 60 °C, and the air receives heat at constant-volume until its temperature reaches 510 °C. Determine: (i) work done per kg of air (ii) change in internal energy per kg of air (iii) the amount of heat added to the system. Take $C_{v,air} = 0.71 \text{ kJ/kgK}$.

[5 Marks]

QUESTION 2

(a) A piston and cylinder machine containing a fluid system has a stirring device as shown in Fig. Q2a

The piston is frictionless, and it is held down against the fluid due to atmospheric pressure of 101. kPa. The stirring device is turned at angular speed of 9500 rev/min, dissipating 1.24 kW power on the fluid. Meanwhile the piston of 0.65 m-diameter moves out 0.6 m. Determine the average torque of the stirring device against the fluid. Find also the work done due to piston movement, indicating its effect on the net work transfer for the system [5 Marks]

(vi) Is it true the assertion: "adiabatic systems are isolated"? Justify your answer: see Ks]

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2. millivoltmeter is calibrated at ice and steam,
here the gas then

mometer reads 80°C

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at 80°C and 80% is

The pressure at the end of

is 1077 kPa . The

is thermometer with

flow process

expansion is 0.7 kJ/m^3 . The gas

@ work transfer

volume at the end

9 the heating process. Take C_p

the heating process, and

in the expansion process, (iii) the

internal energy and $C_p = 0.74 \text{ kJ/kgK}$.

internal energy during the

is a : [10 Mark

QUESTION 4 — : -. des P a

At the beginning of the compression process of an air-standard

compression ratio of 43, the temperature is 300 K and

is 1 MPa

the Diesel cycle operating with

the pressure is 0.4 MPa , The cut-off ratio is

at the end of each process of

pressure, in

Ge

Temperature is

the cycle

Pa, with respect to

is 1 MPa

15 Aarts

point (iii) Saturation point and

E . [4 Mark

the specific internal energy for each of the

: Be 'bar, > 50927 .@7); 330.8°C, $v = 0.01278 \text{ m}^3/\text{kg}$ (iii) p
vps 50Bar, h = sey WIA. Sketch the-steam on @ P-V diagram, clear.

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operation,

(c) Consider ot mestio ped

15,060 Wh refrigeration capacity, Given th

Mi, deter ne the following: (i) the co-efficier

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Paper 3 of 3 — GET 206 Fundamentals of Thermodynamics Handout

Level 200 • First Semester • Handout

This source paper is a scanned document; its pages are reproduced below.

Vapour and Gas Power Cycle ①

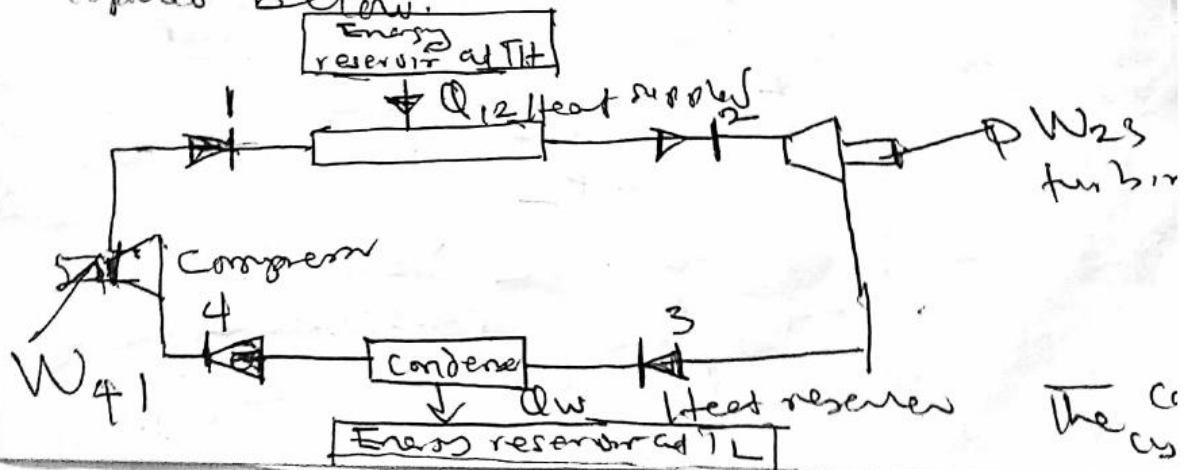
This lecture is covered with the applications of the previous laws in the thermodynamic analysis of cycles used for conversion of heat energy to mechanical work.

The Carnot cycle

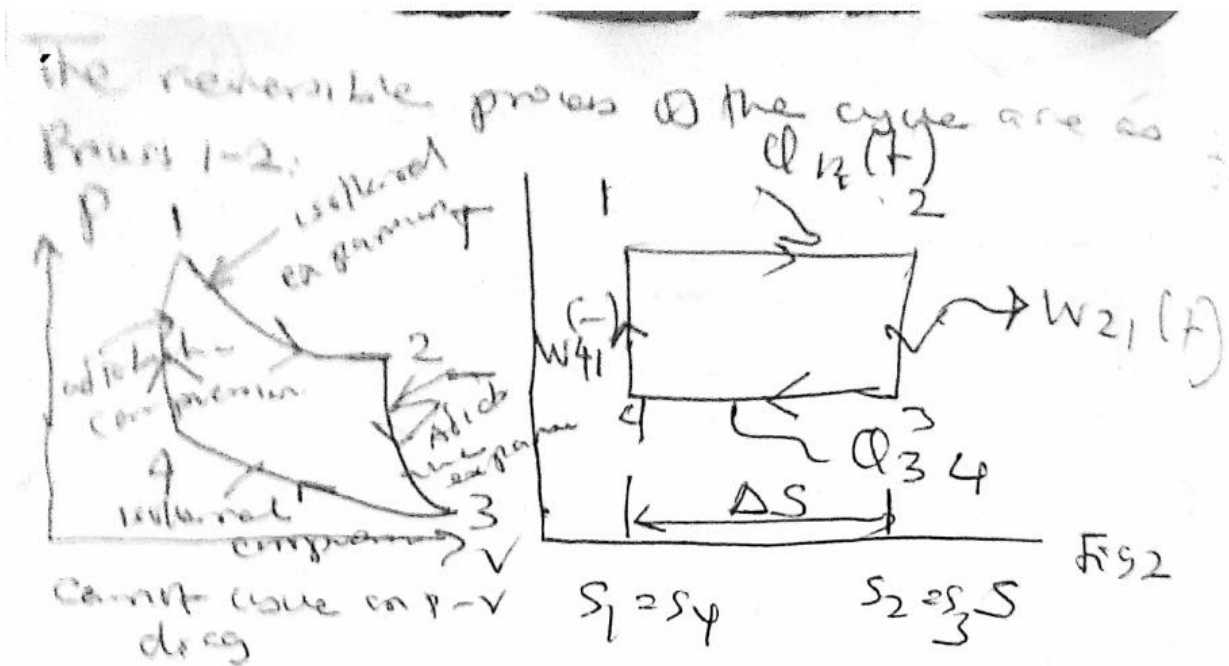
This is a cycle suggested by Sadi Carnot (1796-1837).

It is a totally reversible cycle and provides the maximum theoretical efficiency of a heat engine which cannot be achieved by actual heat engines.

The Carnot cycle consists of two reversible isothermal (constant temperature) processes joined by two reversible adiabatic (isentropic) processes. A diagrammatic sketch of a heat engine working on a Carnot cycle is depicted below.



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Carnot cycle on T-S diagram

Process 1-2: a reversible heat supply to the working fluid at constant temperature T_H and pressure P_H

Process 2-3: a reversible adiabatic ($S_2 = S_3$) expansion of the fluid from high temperature T_H and pressure P_H to low temperature T_L and pressure P_L . During this process work is done by the expanding fluid on the surroundings.

Process 3-4: A reversible heat rejection at constant temperature T_L and pressure P_L

Process 4-1: A reversible adiabatic (isentropic $S_4 = S_1$) compression of the fluid from the temperature T_L and pressure P_L to its original state 1. The compressor does work on the fluid during the process.

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Referring to Fig. 2, and making use of equation
 coefficient of performance (COP) = ratio of the desired
 effect to the work input to achieve this effect

$$COP = \text{desired effect} / \text{work input}$$

Regardless of the working fluid employed in the cycles,
 the heat transfers are:

Heat supplied $Q_{12} = T_H \Delta S$ — (1)

Heat rejected $Q_{34} = T_L \Delta S$ — (2)

Note $\Delta S = S_2 - S_1 = S_2 - S_4$ in eqs (1) & (2)

The Carnot efficiency is given by

$$\eta_{\text{Carnot}} = \frac{\text{net work output}}{\text{heat input}} = \frac{W_{\text{net}}}{Q_{12}} \quad \text{--- (3)}$$

Application of the first law to the cycle yields

$$W_{\text{net}} = Q_{\text{net}}$$

$$W_{\text{net}} = Q_{12} - Q_{24} \quad \text{--- (4)}$$

~~Sub eqn (4) and (2) in equation~~

~~sub eq (1) and (2) in equation~~

sub eq (1) and (2) gives

$$\eta_{\text{Carnot}} = \frac{Q_{12} - Q_{24}}{Q_{12}} = 1 - \frac{Q_{24}}{Q_{12}} \quad \text{--- (5)}$$

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substituting eq. (1) and (2) in eq. (3) we have (4)

$$\eta_{\text{Carnot}} = \frac{1 - T_L \Delta S}{T_H \Delta S}$$

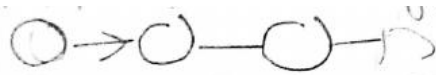
$$\boxed{\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H}} \quad \text{--- (5)}$$

From eq. (5), we can conclude that the thermal efficiency of a heat engine which operates in the Carnot cycle depends only on the temps of the 2 reservoirs b/w which heat exchanges take place. It can be observed that the higher the temp at which heat is supplied the higher the efficiency and the lower the temp at which heat is rejected the higher the efficiency.

problem

A heat engine which operates in a Carnot cycle receives 700 kJ of heat energy at 850°C. The efficiency of the heat engine is 48%. Determine the net work output, the heat rejected and the temperature at which the heat is rejected.

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solution

Heat engine operating on the Carnot cycle

$$\eta_{\text{Carnot}} = 48\% = 0.48$$

$$T_H = 850 + 273 = 1123 \text{ K}, Q_H = Q_{12} = 700 \text{ kJ}$$

= heat input

Find (i) Net work output

(ii) Q_L heat rejected(iii) T_L temperature at which heat is rejectedSolr (i) What $= ?$

$$\eta_{\text{Carnot}} = \frac{W_{\text{net}}}{Q_{12}} = 0.48 \quad \text{--- (1)}$$

$$W_{\text{net}} = 0.48 \times Q_{12} = 0.48 \times 700$$

$$W_{\text{net}} = 336 \text{ kJ}$$

(ii) $Q_L = ?$

Application of the 1st Law to the Carnot cycle gives

$$W_{\text{net}} = Q_H - Q_L$$

$$Q_L = Q_H - W_{\text{net}} = 700 - 336$$

$$Q_L = 364 \text{ kJ}$$

(iii) $T_L = ?$

$$\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H} \quad \text{--- (2)}$$

Comparing eq (1) and (2) gives

$$1 - \frac{T_L}{T_H} = 0.48$$

$$\frac{T_L}{T_H} = 1 - 0.48 = 0.52$$

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$$\frac{T_L}{T_H} = 0.52$$

$$T_L = 0.52 \times T_H = 0.52 \times 1123$$

$$T_L = 584 \text{ K or } 311^\circ\text{C}$$

Impracticability of the Carnot Steam Cycle

- ① The Carnot cycle has the max. theoretical thermal efficiency for a heat engine but the cycle is not realizable for 3 reasons
- ① All the four processes have to be reversible P517B
- ② The heat rejection and absorption have to take place with infinitesimal temp diff.
- ③ For attaining isothermal expansion, the rate must be required to be very low

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Problems on the Carnot cycle & refrigeration

- ① A heat engine working on Carnot cycle ~~operates~~ converts one-fifth of the heat input into work. When the temp of the sink is reduced by 80%, the efficiency gets doubled. Determine the temp of source & sink.

(Ans. Temp of sink, T_2 or $T_L = 320\text{ K}$
Temp of source, $T_1 = 1.25 \times 320 = 400\text{ K}$)

- ② A reversible heat engine delivers 0.6 kW power and rejects heat energy for a reservoir at 300 K at the rate of 24 kJ/min. Calculate the engine efficiency and the temp. of the thermal reservoir supplying heat to the engine.

(Ans. Thermal efficiency, $\eta = 60\%$
 $T_1 = 750\text{ K}$)

(Ans. $\dot{Q}_1 = 215.9736\text{ kJ/min}$)

- ③ An engine mounted on a ship has a thermal efficiency of 80% of that of the corresponding Carnot cycle. The engine receives heat from the sea at 300 K and rejects heat to the atmosphere at 280 K. The work output from the engine is dissipated through an agitator to heat 500 kg of sea water to 355 K. What quantity of heat must be extracted from the sea water to provide the required heating effect. Take specific heat of sea water $c_p = 4.186\text{ kJ/kg K}$.

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Example on the steam tables

Tables of Thermodynamics Properties
The relationship between pressure, volume and temperature and other thermodynamic properties of several pure substances are usually presented in tabular or graphical forms.

Examples

1. Use the steam tables to determine the following:
 - a) the specific internal energy of steam of 90% quality at a pressure of 4.0 bar
 - b) the dryness fraction of steam at 1 bar with a specific enthalpy of 2340 kJ/kg.

Solution

working fluid, steam, $x = 0.90$, $P = 4.0 \text{ bar}$
 $u = ?$ (to find specific internal energy)

$$u = u_f + x(u_g - u_f)$$

The 90% quality indicates that the fluid is in the wet vapour region, hence,

$$u = u_f + x(u_g - u_f)$$

The saturated water/steam table p.54 is consulted to obtain u_f and u_g at P .

$$\text{Hence } u = u_f + x(u_g - u_f)$$

$$605 + 0.90(2554 - 605)$$

$$u = 2359 \text{ kJ/kg}$$

3552 kJ/kg heat

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⑤ Working fluid: steam at $P_2 = 1 \text{ bar}$, $h = 2340 \text{ kJ/kg}$ 011

Find, $x =$ dryness fraction or quality
By requesting for the dryness fraction or quality means that the fluid is in the wet vapour region, so that

$$h = h_f + x h_{fg}$$

from which

$$x = \frac{h - h_f}{h_{fg}}$$

Look at the saturated/water steam table
(p. 44)

at $P_2 = 1 \text{ bar}$, $h_f = 417 \text{ kJ/kg}$

$$h_{fg} = 2258 \text{ kJ/kg} = (h_g - h_f)$$

$$x = \frac{2340 - 417}{2258} = 0.85$$

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Saturated water and Steam, specific enthalpy

P (bar)	P (MPa)	v_g (m ³ /kg)	u_f kJ/kg	u_g kJ/kg	h_f	h_g	s_f	s_g	s_g
1.0			417	417					
4.0			605	2284					

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